GYUSEOK LEE

Ph.D. Student, School of Information Sciences
University of Illinois Urbana-Champaign
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RESEARCH INTERESTS

Recommender Systems · Personalization · Continual Learning · Agentic Search & Memory

EDUCATION

Ph.D., Information Sciences, University of Illinois Urbana-Champaign — Aug 2025 - Present
Advisor: Prof. Jingrui He — *Champaign, IL*
Cumulative GPA: 3.83 / 4.00

M.S., Artificial Intelligence, Pohang University of Science and Technology — Feb 2022 - Feb 2024
Advisor: Prof. Hwanjo Yu — *Pohang, South Korea*
Cumulative GPA: 3.77 / 4.30
Thesis: Collaborative Knowledge Distillation for Continual Learning in Recommender System

B.S., Computer Science / Mathematics and Statistics, Handong Global University — Mar 2015 - Feb 2022
Minor: International and Area Studies — *Pohang, South Korea*
Advisors: Prof. InJung Kim and Prof. Heonjoo Kim
Cumulative GPA: 4.01 / 4.50 & Major GPA: 4.22 / 4.50 (Magna Cum Laude)
Includes 23 months of mandatory military service in the Republic of Korea Air Force

INDUSTRY EXPERIENCE

Applied Scientist Intern, Amazon — Jun 2026 - Sep 2026
Cold-start problem in large-scale recommender systems — *Seattle, WA*
Manager: Shreya Chakrabarti

Medical AI Research Intern, AITRICS — Dec 2021 - Feb 2022
Machine learning for clinical prediction and pathology imaging — *Seoul, South Korea*

1. Two-stage CNN and XGBoost pipeline for thyroid cancer prediction
2. Self-supervised learning for pathology image classification with limited annotations

ACADEMIC EXPERIENCE

Visiting Researcher, Korea University — Apr 2025 - Aug 2025
LLM-generated intents for session-based recommendation [SIGIR 2026] — *Seoul, South Korea*
Host: Prof. SeongKu Kang

Researcher, Pohang University of Science and Technology — Mar 2024 - Aug 2025
Continual sequential recommendation under non-stationary data streams [WSDM 2026] — *Pohang, South Korea*

Host: Prof. Hwanjo Yu

Visiting Researcher, University of Virginia Sep 2023 - Feb 2024
Collaborative diffusion models for recommender systems [WWW 2025] *Charlottesville, VA*
Host: Prof. Jundong Li

Visiting Researcher, Carnegie Mellon University Aug 2022 - Feb 2023
IITP AI Intensive Program, sponsored by the Korean Government *Pittsburgh, PA*

1. Visual clustering for personalized multimodal fashion recommendation (project lead)
2. Multi-agent reinforcement learning for taxi repositioning with attention networks (project lead)
3. Masked language modeling for question answering and question generation
4. Generative model for automatic paper assessment and review scoring

Graduate Research Assistant, Pohang University of Science and Technology Feb 2022 - Feb 2024
Knowledge distillation for continually updated recommender systems [KDD 2024] *Pohang, South Korea*
Advisor: Prof. Hwanjo Yu

Undergraduate Research Intern, Handong Global University Feb 2021 - Dec 2021
Developing DNN partial learning algorithm for on-device learning *Pohang, South Korea*
Supported by ETRI (Electronics and Telecommunications Research Institute)
Advisor: Prof. InJung Kim

PUBLICATIONS

* Equal contribution.

First-Authored (6)

1. SCoRD: Semantic-Assisted Continual Retriever-Reranker Distillation for LLM-Based Recommendation
   Seunghyun Baek*, Gyuseok Lee*, Seunghan Lee, Wonbin Kweon, Dong Wang, SeongKu Kang
   CIKM 2026 | ACM International Conference on Information and Knowledge Management
2. Uncertainty-Aware Variational Reward Factorization via Probabilistic Preference Bases for LLM Personalization
   Gyuseok Lee, Wonbin Kweon, Zhenrui Yue, SeongKu Kang, Jiawei Han, Dong Wang
   COLM 2026 | Conference on Language Modeling
3. SPRINT: Scalable and Predictive Intent Refinement for LLM-Enhanced Session-Based Recommendation
   Gyuseok Lee, Wonbin Kweon, Zhenrui Yue, Yaokun Liu, Yifan Liu, Susik Yoon, Dong Wang, SeongKu Kang
   SIGIR 2026 | ACM SIGIR Conference on Research and Development in Information Retrieval
4. Capturing User Interests from Data Streams for Continual Sequential Recommendation
   Gyuseok Lee, Hyunsik Yoo, Junyoung Hwang, SeongKu Kang, Hwanjo Yu
   WSDM 2026 | ACM International Conference on Web Search and Data Mining
5. Collaborative Diffusion Model for Recommender System
   Gyuseok Lee, Yaochen Zhu, Hwanjo Yu, Yao Zhou, Jundong Li
   WWW 2025 | ACM The Web Conference, Short Paper
6. Continual Collaborative Distillation for Recommender System
   Gyuseok Lee*, SeongKu Kang*, Wonbin Kweon, Hwanjo Yu
   KDD 2024 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining

Co-Authored (2)

7. Topology-Aware Tokenization for Generative Recommendation

Yaokun Liu, Yifan Liu, Zhenrui Yue, Gyuseok Lee, Zelin Li, Ruichen Yao, Dong Wang
RecSys 2026 | ACM Conference on Recommender Systems

8. Harmonic Dataset Distillation for Time Series Forecasting
Seungha Hong, Sanghwan Jang, Wonbin Kweon, Suyeon Kim, Gyuseok Lee, Hwanjo Yu
AAAI 2026 | AAAI Conference on Artificial Intelligence

Under Review (5)

9. Mitigating the Fairness Steering Tax via Adaptive Inference-time Subspace Intervention
Yifan Liu, Yaokun Liu, Ching-Yu Lin, Daniel Yue Zhang, Ruichen Yao, Zelin Li, Gyuseok Lee, Mano Paul, Rahul Gupta, Dong Wang
EACL 2027 | Conference of the European Chapter of the Association for Computational Linguistics

10. SIPO: Unifying Reinforcement Learning with On-Policy Self-Distillation
Zhenrui Yue, Huimin Zeng, Yueqi Wang, Yaokun Liu, Fengran Mo, Jinghan Zhang, Mung Yao Jia, Gyuseok Lee, Yang Zhang, Na Wei, Dong Wang
ICLR 2027 | International Conference on Learning Representations

11. ContextRank: Enhancing Scientific Document Retrieval with Context-Enriched Relations
Wonbin Kweon, Xueqiang Xu, Runchu Tian, Chih-Hsuan Wei, ZhiZheng Wang, Hyunsik Yoo, Gyuseok Lee, SeongKu Kang, Zhiyong Lu, Jiawei Han, Hwanjo Yu
ARR October 2026 | ACL Rolling Review

12. Selective Preference Unlearning for Continual Sequential Recommendation
Dongha Kim, Gyuseok Lee, Suyeon Kim, Jaehyung Lim, Junyoung Kim, SeongKu Kang, Hwanjo Yu
WWW 2027 | The Web Conference

13. DUET: Simulating Interactions for Cold-Start Use Cases at Amazon Prime via Dual Semantic-to-Collaborative Alignment
Gyuseok Lee, Alan Raydan, Zijian Ni, Shreya Chakrabarti, Wonbin Kweon, Jingrui He
RecSys 2027 | ACM Conference on Recommender Systems (To be submitted)

PATENTS

1. Intent Extraction from LLMs for Session-Based Recommendation (KR 10-2026-0106597) — 2026

AWARDS

1. NSF Travel Award ($1,500), ACM WSDM — Feb 2026
2. Facilitating Learning Excellence Award, University of Illinois Urbana-Champaign — Dec 2025

SCHOLARSHIPS

1. Foundation Scholarship ($2,000), OTEFE Foundation, USA — Aug 2026
2. Foundation Scholarship ($1,000), OTEFE Foundation, USA — Aug 2025
3. Foundation Scholarship, GyeongWon Scholarship Foundation, South Korea — Nov 2021
4. Foundation Scholarship, GyeongWon Scholarship Foundation, South Korea — Mar 2021
5. Academic Excellence Scholarship, Handong Global University, South Korea — Mar 2021
6. Academic Excellence Scholarship, Handong Global University, South Korea — Sep 2020
7. Academic Excellence Scholarship, Handong Global University, South Korea — Mar 2020

PROFESSIONAL SERVICE

Reviewer for all entries.

1. ACM Transactions on Information Systems (TOIS) — 2025, 2026

2. Conference on Language Modeling (COLM) 2026
3. ACM SIGIR Conference (SIGIR) 2026
4. ACM The Web Conference (WWW) 2025, 2027

TEACHING

Teaching assistant unless otherwise noted.

1. Foundations of Data Curation, University of Illinois Urbana-Champaign — Fall 2026
2. Reading and Writing Data, University of Illinois Urbana-Champaign — Spring 2026
3. Data Mining, University of Illinois Urbana-Champaign — Fall 2025
4. Artificial Intelligence, SK industry training program (instructor) — Summer 2024, 2025
5. Object-Oriented Programming, Pohang University of Science and Technology — Spring 2023
6. Big Data Analysis with R, Handong Global University — Fall 2021
7. Mathematical Statistics, Handong Global University — Summer 2021
8. Statistical Methodology, Handong Global University — Spring 2021
9. Python Programming, Handong Global University and KMOOC — Spring, Fall 2020